

Expected Values

Quiz, 10 questions

✓ **Congratulations! You passed!**

Next Item



1 / 1
points

1.

Let X be a multivariate random vector and A and b be a constant matrix and vector respectively. What is true about $E[AX + b]$? (Assume the dimensions agree, check all that apply.)



1 / 1
points

2.

Let $X_1 \dots X_n$ all be $p \times 1$ vectors, not necessarily independent. Let $E[X_i] = \mu_i$. Let $\bar{X}$ be the elementwise average of the $\{X_i\}$ (i.e. $\frac{1}{n} \sum_{i=1}^n X_i$). What is $E[\bar{X}]$. (Check all that apply.)



1 / 1
points

3.

$Var(AX + b)$ is equal to what (check all that apply)?



1 / 1
points

4.

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Let $C(X, Y) = E[(X - \mu_x)(Y - \mu_y)']$ be the multivariate covariance operator for (not necessarily equal dimensional) vectors X and Y with means $E[X] = \mu_x$ and $E[Y] = \mu_y$. What is $C(X, Y)$ equal to (check all that apply)?



1 / 1
points

5.

Let $X_1 \dots X_n$ all be $p \times 1$ independent vectors. Let $E[X_i] = \mu_i$ and $Var(X_i) = \Sigma$. Let $\bar{X}$ be the elementwise average of the $\{X_i\}$ (i.e. $\frac{1}{n} \sum_{i=1}^n X_i$). What is $Var(\bar{X})$ (check all that apply)?

0.75 / 1
points

6.

Let X_1 and X_2 be equi-dimensional random vectors and let Y be another random vector. What is $C(X_1 + X_2, Y)$ equal to (check all that apply)?

0.33 / 1
points

7.

What is the transpose of $C(X, Y)$?

0.80 / 1
points

8.

Let A and B be constant matrices of the relevant dimensions. What is $C(AX, BY)$? (Check all that apply.)

0.25 / 1
points

9.

Let X be a random vector with $E[X] = \mu$ and $Var(X) = \Sigma$. What is $E[X'X]$ equal to? (Check all that apply.)

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1 / 1
points

10.

Let x and β be a fixed vectors and Y be a random vector, so that $E[Y] = x\beta$ and $Var(Y) = \Sigma$. Let $\hat{\beta}(w) = (x'wx)^{-1}x'wY$ where w is a square, symmetric, invertible matrix. What can be said about this estimator? (Check all that apply.)

